End Semester Exams. 21st Nov.25

INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, November-December, 2025

Degree: **B.Tech.** Semester: **First**

Stream: **CSBS/CSE/CSE(AI & ML)/CSE(AI)/ CSE(IOT)/ CSE(IOT, CSBT)/CSE(Cyber Security)/ CSIT/CST/CSE(Data Science)/CSE(Networks)/ ECE/EEE/EE/ME/IT/Robotics & AI**

Paper Code: **ESCCS102**

Paper Name: **Programming for Problem Solving using C**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. Which one among the following data types takes the least space in memory, and which one takes the most? **2**
char *, char **, int *, float ***, long double

2. Given float z = 0.5625e2;, what is the smallest positive integer that should be multiplied with z so that the result is an integer? **2**

3. Write the value of the expression for a = 5, b = 6, c = 11: **2**
a = = ((2*a*b - 22 < 3*c) && (a + b = = c))

4. Write the value of the expression: (a & b) & (a & b) where a = 3, b = 4, c = 5. **2**

5. How many element-to-element comparisons are sufficient to find the median among 5 distinct numbers? **2**

6. How many multiplications are enough to compute the value of x^128 (i.e., x^{128}), where x is a real number in [0, 1] (without using the math library)? **2**

7. Which runs faster for computing the n-th Fibonacci term when n is a positive **2**

integer less than 13: a recursive function or an iterative function? Why?

8. For the following mathematical expressions, write their corresponding C expressions. Assume x > 0 and is declared as a variable of type double. You have to use functions from math.h. 2

i) log(x+5) + (2e) ^x

ii) (sin (x) + cos (x)) ^x

9. How many bits are needed to store each letter in "India" and how many bits are needed to store "IEM" as a valid string with a null terminator? 2

10. Why is the pre-increment operator (++x) sometimes preferred over the post-increment operator (x++) in C? 2

11. How does the switch-case statement execute multiple cases when no break is present, and how can this behavior be controlled? 2

12. Which statement is used inside a loop to skip the current iteration and continue with the next iteration? 2

13. Using suitable examples, explain how the concept of pass by value differs from pass by reference in C functions. 2

14. In C, why is it recommended to use sizeof operator when allocating memory dynamically? 2

15. What happens if a function in C does not have a return statement, and what is the default behavior? 2

Part - B

Attempt any 6 questions out of 9 questions

Each question carries 5 Marks (5 × 6)

16. Explain the difference between malloc() and realloc() in dynamic memory allocation. 5

17. Write a C program to calculate the GCD of two numbers using recursion. 5

18. Write a program that counts the number of vowels in a string. 5

19. Write a C program to check if a number is a palindrome using only arithmetic operations. 5

20. Explain the difference between strcmp() and strncmp() functions in C. Provide examples. 5

21. Write a function that prepares a sorted array containing the common elements of two sorted arrays. 5

22. Write a program to count the frequency of each character in a string using an array. 5

23. Write a program that prints all prime numbers less than n. 5

24. Write a function that finds the second largest element in an array. 5

Part - C

Attempt any 5 questions out of 8 questions

Each question carries 10 Marks (10 × 5)

25. The coordinates (x and y) of 10 different points are entered through the keyboard. Write a C program that prints the points in non-decreasing order of their distances from the 1st point. 10

26. Write an iterative function that takes a positive integer from the user and returns its binary equivalent. The main program should call this function and print the result. 10

27. Write a C program to compute the multiplication of two matrices of size 5-by-5. 10

28. Write a C program to implement a game on a 2D binary array a[0..9][0..9] of size 10-by-10. The player (user) will start from start = a[0][0] and will try to reach end = a[9][9] by moving either horizontally or vertically at each step only from a 0-cell to a 0-cell. A cell with 1 is blocked. Assume that there is a unique path from start to end, and there is no other path. The user cannot visit a cell more than once. Her score will be 10/10 if she can reach within 10 seconds. Assume that there is an inbuilt clock function to measure the time. 10

29. Write a C program to find and print all prime numbers in a given range, which are of the form 4k+1, where k is a natural number. 10

30. Write a C program to compute the summation (S) of the following series, where the number of terms (N) is provided as user input: 10
S = 1 + 101 + 101 + 1001 + ... up to N terms.

31. Write a C program to read a text file and remove all the punctuations. Assume that the file contains only letters, digits, spaces, and punctuations. 10

32. Define a structure named vertex that will contain the coordinates of a 2D point. Define another structure named triangle to hold three vertices. Write a program to create an array of 10 triangles, accept data for them, and then 10

print a/the triangle with highest perimeter.

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